

Code and analyses for *National income inequality predicts cultural variation in mouth to mouth kissing*

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Contents

Preliminaries	1
Import packages	1
Loading data	1
Factors	2
Rescale variable GDP	2
Excluding participants	2
Individual differences in attitudes and behaviours related to kissing and intimacy	2
Mixed model	2
Follow-up tests	2
Multilevel models	4
Models Importance of kissing at initial phase	4
Models Importance of kissing at established phase	7
Models frequency of kissing	11
Models frequency of hugging	14
Models sex frequency	17
Models kissing satisfaction	20
Models Hugging satisfaction	24
Models Sex satisfaction	27
Principal Component Analysis	31
PCA on the determinants of a good kiss	31
Varimax rotation	35
Excluding participants from the new dataset	36
Models with variables from the PCA	36
Models dim 1 (“technique”) of a good kiss	36
Models dim 2 (“sensory”) of a good kiss	39
Participant sex and importance of the sensory component of a good kiss	42

This R Markdown document contains the code and analyses for Watkins, C.D., Leongómez, J.D., Bovet, J., Zelazniewicz, A., Korbmacher, M., Corrêa Varella, M.A., Fernandez, A.M., Wagstaff, D., & Bolgan S. (2019). *National income inequality predicts cultural variation in mouth to mouth kissings*. Manuscript in preparation.

Preliminaries

Import packages

Loading data

```
load("Kissing_submitted_data_clean.RData")
```

Factors

```
db_sample$Sex <- as.factor(as.character(db_sample$Sex))
db_sample$RelStatus <- as.factor(as.character(db_sample$RelStatus))
```

Rescale variable GDP

```
## k$ instead of $
db_sample$GDP <- db_sample$GDP/1000
```

Excluding participants

```
db <- subset(db_sample, db_sample$Ex_Cross==0)
db$Country_Born <- droplevels(db$Country_Born)
db$Country_Res <- droplevels(db$Country_Res)
```

Individual differences in attitudes and behaviours related to kissing and intimacy

Mixed model

```
#Create database with only relevant columns, and *melting* kiss importance into one new
#column (Kiss_Imp), and one factor (Rel_phase)
db_Kiss_Imp <- melt(db, id.vars = c(1, 5, 7, 21), measure.vars = 27:28,
                    variable.name = "Rel_phase",
                    value.name = "Kiss_Imp")
db_Kiss_Imp$ID <- as.factor(db_Kiss_Imp$ID)
m_Kiss_phases <- lmer(Kiss_Imp ~ Rel_phase + Sex + SRA + Age +
                     Rel_phase:Sex + Rel_phase:SRA + Rel_phase:Age +
                     (1 | ID), data = db_Kiss_Imp)
anova(m_Kiss_phases, type = 3)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##              Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
## Rel_phase      4563.0   4563.0      1 1906.0  18.8364 1.499e-05 ***
## Sex            1223.5   1223.5      1 1905.3   5.0505 0.024733 *
## SRA            1898.8   1898.8      1 1907.1   7.8384 0.005166 **
## Age           12136.3  12136.3      1 1904.2  50.0992 2.047e-12 ***
## Rel_phase:Sex    111.0    111.0      1 1904.3   0.4581 0.498578
## Rel_phase:SRA     22.9     22.9      1 1906.1   0.0945 0.758561
## Rel_phase:Age   4959.6   4959.6      1 1903.2  20.4735 6.420e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Follow-up tests

Importance of kissing by relationship phase

```
t.test(Kiss_Imp ~ Rel_phase, db_Kiss_Imp, conf.int=TRUE)
```

```
##
## Welch Two Sample t-test
```

```
##
## data: Kiss_Imp by Rel_phase
## t = -4.7615, df = 3669.8, p-value = 1.996e-06
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -4.744784 -1.977005
## sample estimates:
## mean in group Kiss_Imp_Initial      mean in group Kiss_Imp_Est
##                81.03848                84.39938
group.CI(Kiss_Imp ~ Rel_phase, data = db_Kiss_Imp, ci = 0.95)

##          Rel_phase Kiss_Imp.upper Kiss_Imp.mean Kiss_Imp.lower
## 1 Kiss_Imp_Initial      82.11872      81.03848      79.95825
## 2 Kiss_Imp_Est         85.26505      84.39938      83.53370
cohen.d(Kiss_Imp ~ Rel_phase, data = db_Kiss_Imp)

##
## Cohen's d
##
## d estimate: -0.1535572 (negligible)
## 95 percent confidence interval:
##      lower      upper
## -0.21687839 -0.09023594
```

Importance of kissing by age

```
cor.test(db_Kiss_Imp$Kiss_Imp, db_Kiss_Imp$Age, conf.level = 0.95)

##
## Pearson's product-moment correlation
##
## data: db_Kiss_Imp$Kiss_Imp and db_Kiss_Imp$Age
## t = 8.3291, df = 3830, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.1021507 0.1643524
## sample estimates:
##      cor
## 0.1333829
```

Importance of kissing by self-rated attractiveness

```
cor.test(db_Kiss_Imp$Kiss_Imp, db_Kiss_Imp$SRA, conf.level = 0.95)

##
## Pearson's product-moment correlation
##
## data: db_Kiss_Imp$Kiss_Imp and db_Kiss_Imp$SRA
## t = 3.3434, df = 3836, p-value = 0.0008354
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.02230331 0.08539730
## sample estimates:
##      cor
```

```
## 0.05390411
```

Importance of kissing by sex

```
t.test(Kiss_Imp ~ Sex, db_Kiss_Imp, conf.int=TRUE)
```

```
##
## Welch Two Sample t-test
##
## data: Kiss_Imp by Sex
## t = -1.6941, df = 1722.8, p-value = 0.09044
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -3.0136557 0.2203602
## sample estimates:
## mean in group 1 mean in group 2
##      81.69523      83.09188
```

```
group.CI(Kiss_Imp ~ Sex, data = db_Kiss_Imp, ci = 0.95)
```

```
## Sex Kiss_Imp.upper Kiss_Imp.mean Kiss_Imp.lower
## 1 1      83.10347      81.69523      80.28699
## 2 2      83.88756      83.09188      82.29619
```

```
cohen.d(Kiss_Imp ~ Sex, data = db_Kiss_Imp)
```

```
##
## Cohen's d
##
## d estimate: -0.06364984 (negligible)
## 95 percent confidence interval:
##      lower      upper
## -0.135122973 0.007823291
```

Multilevel models

Models Importance of kissing at initial phase

```
m_Kiss_Imp_Initial_1 <- lmer(Kiss_Imp_Initial ~ HPP_9 + (1|Country_Born), data=db)
summary(m_Kiss_Imp_Initial_1)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Kiss_Imp_Initial ~ HPP_9 + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 17600.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.6325 -0.3785  0.3276  0.6792  1.7596
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## Country_Born (Intercept) 39.75    6.305
```

```

## Residual                    547.09    23.390
## Number of obs: 1923, groups: Country_Born, 13
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   78.581      1.931 11.085  40.692 2.01e-13 ***
## HPP_9         -5.431      2.481 11.033  -2.189   0.051 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## HPP_9 0.231

m_Kiss_Imp_Initial_2 <- lmer(Kiss_Imp_Initial ~ HPP_9 + GINI + GDP + (1|Country_Born),
                             data=db)
summary(m_Kiss_Imp_Initial_2)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Kiss_Imp_Initial ~ HPP_9 + GINI + GDP + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 17602.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.6364 -0.3658  0.3285  0.6793  1.7742
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## Country_Born (Intercept) 49.32    7.023
## Residual              547.09   23.390
## Number of obs: 1923, groups: Country_Born, 13
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) 82.36261   13.17784 10.07149   6.250 9.21e-05 ***
## HPP_9        -5.27464    4.60346  9.03243  -1.146   0.281
## GINI         -0.07956    0.25586  9.52070  -0.311   0.763
## GDP          -0.02328    0.21281  9.60737  -0.109   0.915
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) HPP_9  GINI
## HPP_9 -0.169
## GINI  -0.878 -0.184
## GDP   -0.659  0.711  0.255

m_Kiss_Imp_Initial_3 <- lmer(Kiss_Imp_Initial ~ HPP_9 + GINI + GDP + Sex + Age + SRA +
                             RelStatus + (1|Country_Born), data=db)
summary(m_Kiss_Imp_Initial_3)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [

```

```
## lmerModLmerTest]
## Formula:
## Kiss_Imp_Initial ~ HPP_9 + GINI + GDP + Sex + Age + SRA + RelStatus +
## (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 17330.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.2768 -0.3532  0.3044  0.6713  2.0238
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## Country_Born (Intercept) 63.56   7.973
## Residual                522.51  22.858
## Number of obs: 1903, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  63.29020   14.92907   10.48970   4.239  0.00154 **
## HPP_9        -5.45674    5.15941    9.01652  -1.058  0.31774
## GINI         -0.08099    0.28580    9.37101  -0.283  0.78305
## GDP          -0.04368    0.23772    9.45177  -0.184  0.85809
## Sex2          1.55688    1.20752  1889.92086   1.289  0.19745
## Age           0.33345    0.04877  1894.04847   6.837 1.09e-11 ***
## SRA           2.06159    0.44695  1894.98591   4.613 4.24e-06 ***
## RelStatus2   -2.73401    1.23828  1887.25231  -2.208  0.02737 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) HPP_9  GINI   GDP    Sex2   Age    SRA
## HPP_9         -0.169
## GINI          -0.861 -0.185
## GDP           -0.646  0.714  0.248
## Sex2          -0.084  0.002  0.013 -0.001
## Age           -0.084 -0.005 -0.011 -0.022  0.130
## SRA           -0.146  0.001  0.003  0.017  0.005 -0.045
## RelStatus2    -0.024 -0.004 -0.026 -0.009  0.036  0.100  0.091
```

Summary Table Importance of kissing at initial phase

```
class(m_Kiss_Imp_Initial_1) <- "lmerMod"
class(m_Kiss_Imp_Initial_2) <- "lmerMod"
class(m_Kiss_Imp_Initial_3) <- "lmerMod"
stargazer(m_Kiss_Imp_Initial_1, m_Kiss_Imp_Initial_2, m_Kiss_Imp_Initial_3, type = "text",
          digits = 2, star.cutoffs = c(0.05, 0.01, 0.001), report = ('vc*stp'))
```

```
##
## =====
##              Dependent variable:
##              -----
##              Kiss_Imp_Initial
```

```

##              (1)      (2)      (3)
## -----
## HPP_9          -5.43*    -5.27    -5.46
##              (2.48)    (4.60)    (5.16)
##              t = -2.19  t = -1.15  t = -1.06
##              p = 0.03   p = 0.26   p = 0.30
##
## GINI           -0.08     -0.08
##              (0.26)    (0.29)
##              t = -0.31  t = -0.28
##              p = 0.76   p = 0.78
##
## GDP            -0.02     -0.04
##              (0.21)    (0.24)
##              t = -0.11  t = -0.18
##              p = 0.92   p = 0.86
##
## Sex2           1.56
##              (1.21)
##              t = 1.29
##              p = 0.20
##
## Age            0.33***
##              (0.05)
##              t = 6.84
##              p = 0.00
##
## SRA            2.06***
##              (0.45)
##              t = 4.61
##              p = 0.0000
##
## RelStatus2     -2.73*
##              (1.24)
##              t = -2.21
##              p = 0.03
##
## Constant       78.58***  82.36***  63.29***
##              (1.93)   (13.18)  (14.93)
##              t = 40.69  t = 6.25  t = 4.24
##              p = 0.00   p = 0.00   p = 0.0001
## -----
## Observations    1,923    1,923    1,903
## Log Likelihood  -8,800.22 -8,801.37 -8,665.21
## Akaike Inf. Crit. 17,608.44 17,614.75 17,350.42
## Bayesian Inf. Crit. 17,630.69 17,648.12 17,405.93
## =====
## Note:           *p<0.05; **p<0.01; ***p<0.001

```

Models Importance of kissing at established phase

```
m_Kiss_Imp_Est_1 <- lmer(Kiss_Imp_Est ~ HPP_9 + (1|Country_Born), data=db)
summary(m_Kiss_Imp_Est_1)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Kiss_Imp_Est ~ HPP_9 + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 16830.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.4110 -0.4432  0.2941  0.7110  1.1206
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## Country_Born (Intercept)    9.189    3.031
## Residual                  368.150   19.187
## Number of obs: 1923, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  83.1983    1.0347 10.5117  80.411 5.24e-16 ***
## HPP_9        -0.4089    1.3260 10.2453  -0.308  0.764
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## HPP_9 0.238
```

```
m_Kiss_Imp_Est_2 <- lmer(Kiss_Imp_Est ~ HPP_9 + GINI + GDP + (1|Country_Born), data=db)
summary(m_Kiss_Imp_Est_2)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Kiss_Imp_Est ~ HPP_9 + GINI + GDP + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 16833.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.4330 -0.4531  0.2971  0.7112  1.1336
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## Country_Born (Intercept)    9.152    3.025
## Residual                  368.170   19.188
## Number of obs: 1923, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  74.57154    6.70793 11.68833  11.117 1.45e-07 ***
```



```

## HPP_9      -0.71842    2.23167  8.47045  -0.322    0.755
## GINI       0.17578    0.12721 10.02161   1.382    0.197
## GDP        0.05813    0.10631 10.40760   0.547    0.596
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) HPP_9  GINI
## HPP_9 -0.160
## GINI  -0.889 -0.177
## GDP   -0.681  0.695  0.303

m_Kiss_Imp_Est_3 <- lmer(Kiss_Imp_Est ~ HPP_9 + GINI + GDP + Sex + Age + SRA + RelStatus +
  (1|Country_Born), data=db)
summary(m_Kiss_Imp_Est_3)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## Kiss_Imp_Est ~ HPP_9 + GINI + GDP + Sex + Age + SRA + RelStatus +
##      (1 | Country_Born)
##      Data: db
##
## REML criterion at convergence: 16615.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.5110 -0.4540  0.3226  0.7296  1.3574
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
## Country_Born (Intercept) 10.92    3.305
## Residual                360.31   18.982
## Number of obs: 1903, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  63.88747    7.44640   13.33167   8.580 8.59e-07 ***
## HPP_9        -0.65214    2.38204    8.26721  -0.274 0.790975
## GINI         0.18494    0.13495    9.52286   1.370 0.201973
## GDP          0.05275    0.11283    9.86952   0.468 0.650271
## Sex2         1.46302    1.00229  1894.43850   1.460 0.144544
## Age          0.13469    0.04033  1869.31370   3.340 0.000855 ***
## SRA          1.27718    0.37002  1849.39338   3.452 0.000570 ***
## RelStatus2   -1.24026    1.02825  1889.70419  -1.206 0.227893
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) HPP_9  GINI  GDP  Sex2  Age  SRA
## HPP_9      -0.156
## GINI       -0.843 -0.176
## GDP        -0.645  0.699  0.295
## Sex2       -0.139  0.007  0.020 -0.001
## Age        -0.141 -0.005 -0.018 -0.037  0.131

```

```
## SRA          -0.244  0.000  0.008  0.030  0.008 -0.044
## RelStatus2 -0.040 -0.008 -0.046 -0.017  0.035  0.103  0.089
```

Summary Table Importance of kissing at established phase

```
class(m_Kiss_Imp_Est_1) <- "lmerMod"
class(m_Kiss_Imp_Est_2) <- "lmerMod"
class(m_Kiss_Imp_Est_3) <- "lmerMod"
stargazer(m_Kiss_Imp_Est_1, m_Kiss_Imp_Est_2, m_Kiss_Imp_Est_3, type = "text", digits = 2,
          star.cutoffs = c(0.05, 0.01, 0.001),report=('vc*stp'))
```

```
##
## =====
##                               Dependent variable:
##                               -----
##                               Kiss_Imp_Est
##                               (1)      (2)      (3)
## -----
## HPP_9                -0.41      -0.72      -0.65
##                      (1.33)     (2.23)     (2.38)
##                      t = -0.31  t = -0.32  t = -0.27
##                      p = 0.76   p = 0.75   p = 0.79
##
## GINI                  0.18       0.18
##                      (0.13)     (0.13)
##                      t = 1.38   t = 1.37
##                      p = 0.17   p = 0.18
##
## GDP                   0.06       0.05
##                      (0.11)     (0.11)
##                      t = 0.55   t = 0.47
##                      p = 0.59   p = 0.65
##
## Sex2                  1.46
##                      (1.00)
##                      t = 1.46
##                      p = 0.15
##
## Age                   0.13***
##                      (0.04)
##                      t = 3.34
##                      p = 0.001
##
## SRA                   1.28***
##                      (0.37)
##                      t = 3.45
##                      p = 0.001
##
## RelStatus2           -1.24
##                      (1.03)
##                      t = -1.21
##                      p = 0.23
##
## Constant             83.20***   74.57***   63.89***
```

```
##              (1.03)      (6.71)      (7.45)
##              t = 80.41   t = 11.12   t = 8.58
##              p = 0.00    p = 0.00    p = 0.00
##
## -----
## Observations          1,923        1,923        1,903
## Log Likelihood        -8,415.11   -8,416.66   -8,307.80
## Akaike Inf. Crit.    16,838.22   16,845.32   16,635.61
## Bayesian Inf. Crit. 16,860.47   16,878.69   16,691.12
## =====
## Note:                  *p<0.05; **p<0.01; ***p<0.001
```

Models frequency of kissing

```
m_Kiss_Freq_1 <- lmer(Kiss_Freq ~ HPP_9 + (1|Country_Born), data=db)
summary(m_Kiss_Freq_1)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Kiss_Freq ~ HPP_9 + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 16965.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.9404 -0.5154  0.2101  0.9525  1.3183
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## Country_Born (Intercept)    8.926     2.988
## Residual                    417.699    20.438
## Number of obs: 1911, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)   77.2981     1.0449 10.6580  73.979 8.5e-16 ***
## HPP_9         -0.2087     1.3384 10.3289  -0.156  0.879
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## HPP_9 0.240
```

```
m_Kiss_Freq_2 <- lmer(Kiss_Freq ~ HPP_9 + GINI + GDP + (1|Country_Born), data=db)
summary(m_Kiss_Freq_2)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Kiss_Freq ~ HPP_9 + GINI + GDP + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 16961
##
```

```

## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.9324 -0.5055  0.2285  0.8956  1.3194
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
## Country_Born (Intercept)  3.263   1.806
## Residual                417.254  20.427
## Number of obs: 1911, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  64.870163   5.434405  25.247951  11.937 7.03e-12 ***
## HPP_9        -2.931227   1.680493  12.723987  -1.744  0.10520
## GINI         0.321938   0.099885  18.935937   3.223  0.00449 **
## GDP         -0.008696   0.084362  21.088303  -0.103  0.91887
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) HPP_9  GINI
## HPP_9 -0.156
## GINI  -0.903 -0.160
## GDP   -0.710  0.679  0.368

m_Kiss_Freq_3 <- lmer(Kiss_Freq ~ HPP_9 + GINI + GDP + Sex + Age + SRA + RelStatus +
  (1|Country_Born), data=db)
summary(m_Kiss_Freq_3)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Kiss_Freq ~ HPP_9 + GINI + GDP + Sex + Age + SRA + RelStatus +
##      (1 | Country_Born)
##      Data: db
##
## REML criterion at convergence: 16749.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.9996 -0.5364  0.1993  0.8139  1.7649
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
## Country_Born (Intercept)  4.556   2.134
## Residual                406.448  20.161
## Number of obs: 1893, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  56.35261   6.27396  28.33951   8.982 8.74e-10 ***
## HPP_9        -2.93631   1.82999  11.50528  -1.605  0.13567
## GINI         0.37143   0.10707  15.76772   3.469  0.00322 **
## GDP         0.01250   0.09031  17.18466   0.138  0.89154
## Sex2        -0.49241   1.06779 1884.13447  -0.461  0.64474
## Age         -0.01497   0.04266 1695.96554  -0.351  0.72563

```

```
## SRA          1.77752    0.39269 1718.04233    4.527 6.41e-06 ***
## RelStatus2   -4.55024    1.09637 1884.02067   -4.150 3.47e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) HPP_9  GINI   GDP   Sex2   Age   SRA
## HPP_9      -0.149
## GINI        -0.829 -0.161
## GDP         -0.648  0.684  0.350
## Sex2        -0.173  0.015  0.025  0.000
## Age         -0.182  0.004 -0.020 -0.046  0.133
## SRA         -0.311 -0.006  0.019  0.040  0.004 -0.044
## RelStatus2 -0.048 -0.015 -0.062 -0.024  0.034  0.103  0.086
```

Summary Table frequency of kissing

```
class(m_Kiss_Freq_1) <- "lmerMod"
class(m_Kiss_Freq_2) <- "lmerMod"
class(m_Kiss_Freq_3) <- "lmerMod"
stargazer(m_Kiss_Freq_1, m_Kiss_Freq_2, m_Kiss_Freq_3, type = "text", digits = 2,
          star.cutoffs = c(0.05, 0.01, 0.001),report=('vc*stp'))
```

```
##
## =====
##                               Dependent variable:
##                               -----
##                               Kiss_Freq
##                               (1)      (2)      (3)
## -----
## HPP_9          -0.21      -2.93      -2.94
##                (1.34)     (1.68)     (1.83)
##                t = -0.16  t = -1.74  t = -1.60
##                p = 0.88  p = 0.09   p = 0.11
##
## GINI           0.32**     0.37***
##                (0.10)     (0.11)
##                t = 3.22   t = 3.47
##                p = 0.002  p = 0.001
##
## GDP            -0.01      0.01
##                (0.08)     (0.09)
##                t = -0.10  t = 0.14
##                p = 0.92   p = 0.89
##
## Sex2           -0.49
##                (1.07)
##                t = -0.46
##                p = 0.65
##
## Age            -0.01
##                (0.04)
##                t = -0.35
##                p = 0.73
```

```
##
## SRA                                1.78***
##                                (0.39)
##                                t = 4.53
##                                p = 0.0000
##
## RelStatus2                        -4.55***
##                                (1.10)
##                                t = -4.15
##                                p = 0.0001
##
## Constant          77.30***  64.87***  56.35***
##                   (1.04)   (5.43)   (6.27)
##                   t = 73.98 t = 11.94 t = 8.98
##                   p = 0.00  p = 0.00  p = 0.00
##
## -----
## Observations      1,911      1,911      1,893
## Log Likelihood    -8,482.54 -8,480.50 -8,374.88
## Akaike Inf. Crit. 16,973.09 16,973.00 16,769.76
## Bayesian Inf. Crit. 16,995.31 17,006.33 16,825.22
## =====
## Note:                *p<0.05; **p<0.01; ***p<0.001
```

Models frequency of hugging

```
m_Hug_Freq_1 <- lmer(Hug_Freq ~ HPP_9 + (1|Country_Born), data=db)
summary(m_Hug_Freq_1)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Hug_Freq ~ HPP_9 + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 17252.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.5114 -0.4525  0.2928  0.7982  1.2643
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## Country_Born (Intercept) 20.47   4.524
## Residual              507.86  22.536
## Number of obs: 1901, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)   78.904      1.458 10.041  54.133 1.02e-13 ***
## HPP_9         -2.145      1.871  9.919  -1.146   0.279
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
```

```
##          (Intr)
## HPP_9 0.235

m_Hug_Freq_2 <- lmer(Hug_Freq ~ HPP_9 + GINI + GDP + (1|Country_Born), data=db)
summary(m_Hug_Freq_2)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Hug_Freq ~ HPP_9 + GINI + GDP + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 17247.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.5201 -0.4593  0.2950  0.8225  1.2720
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## Country_Born (Intercept)    7.646    2.765
## Residual                  507.823   22.535
## Number of obs: 1901, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  83.4926     6.8755 12.9229  12.144 1.93e-08 ***
## HPP_9        -8.2649     2.2201  8.1342  -3.723 0.00568 **
## GINI          0.1153     0.1287 10.4660   0.896 0.39043
## GDP         -0.3162     0.1080 11.1495  -2.929 0.01353 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) HPP_9  GINI
## HPP_9 -0.158
## GINI  -0.895 -0.170
## GDP   -0.693 0.687 0.331

m_Hug_Freq_3 <- lmer(Hug_Freq ~ HPP_9 + GINI + GDP + Sex + Age + SRA + RelStatus +
(1|Country_Born), data=db)
summary(m_Hug_Freq_3)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Hug_Freq ~ HPP_9 + GINI + GDP + Sex + Age + SRA + RelStatus +
## (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 17040.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.6179 -0.4660  0.2901  0.7989  1.4683
##
## Random effects:
```

```
## Groups      Name      Variance Std.Dev.
## Country_Born (Intercept)  8.809  2.968
## Residual      499.356  22.346
## Number of obs: 1882, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  82.80213    7.63274   15.67166  10.848 1.08e-08 ***
## HPP_9        -8.37423    2.32457    7.79606  -3.602  0.00726 **
## GINI          0.16518    0.13387    9.76270   1.234  0.24612
## GDP         -0.30289    0.11242   10.37063  -2.694  0.02187 *
## Sex2          2.11901    1.18716  1873.92621   1.785  0.07443 .
## Age         -0.08969    0.04755  1764.44564  -1.886  0.05943 .
## SRA           0.26147    0.43925  1751.82663   0.595  0.55174
## RelStatus2   -5.84688    1.21810  1870.45845  -4.800 1.71e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) HPP_9  GINI   GDP    Sex2   Age    SRA
## HPP_9        -0.151
## GINI          -0.835 -0.168
## GDP           -0.646  0.690  0.324
## Sex2          -0.160  0.011  0.022  0.000
## Age           -0.162 -0.002 -0.021 -0.045  0.133
## SRA           -0.286 -0.004  0.015  0.036  0.008 -0.045
## RelStatus2   -0.044 -0.014 -0.055 -0.023  0.034  0.102  0.089
```

Summary Table frequency of hugging

```
class(m_Hug_Freq_1) <- "lmerMod"
class(m_Hug_Freq_2) <- "lmerMod"
class(m_Hug_Freq_3) <- "lmerMod"
stargazer(m_Hug_Freq_1, m_Hug_Freq_2, m_Hug_Freq_3, type = "text", digits = 2,
          star.cutoffs = c(0.05, 0.01, 0.001), report = ('vc*stp'))
```

```
##
## =====
##              Dependent variable:
##              -----
##              Hug_Freq
##              (1)      (2)      (3)
## -----
## HPP_9          -2.14    -8.26***  -8.37***
##              (1.87)    (2.22)    (2.32)
##              t = -1.15 t = -3.72 t = -3.60
##              p = 0.26  p = 0.0002 p = 0.0004
##
## GINI           0.12     0.17
##              (0.13)    (0.13)
##              t = 0.90  t = 1.23
##              p = 0.38  p = 0.22
##
## GDP           -0.32**   -0.30**
```



```

##              (0.11)      (0.11)
##              t = -2.93  t = -2.69
##              p = 0.004  p = 0.01
##
## Sex2              2.12
##              (1.19)
##              t = 1.78
##              p = 0.08
##
## Age              -0.09
##              (0.05)
##              t = -1.89
##              p = 0.06
##
## SRA              0.26
##              (0.44)
##              t = 0.60
##              p = 0.56
##
## RelStatus2      -5.85***
##              (1.22)
##              t = -4.80
##              p = 0.0000
##
## Constant        78.90***  83.49***  82.80***
##              (1.46)   (6.88)   (7.63)
##              t = 54.13 t = 12.14 t = 10.85
##              p = 0.00  p = 0.00  p = 0.00
##
## -----
## Observations      1,901    1,901    1,882
## Log Likelihood    -8,626.26 -8,623.96 -8,520.25
## Akaike Inf. Crit.  17,260.53 17,259.93 17,060.51
## Bayesian Inf. Crit. 17,282.73 17,293.23 17,115.91
## =====
## Note:              *p<0.05; **p<0.01; ***p<0.001

```

Models sex frequency

```

m_Sex_Freq_1 <- lmer(Sex_Freq ~ HPP_9 + (1|Country_Born), data=db)
summary(m_Sex_Freq_1)

```

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Sex_Freq ~ HPP_9 + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 17195.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.8511 -0.6237  0.1301  0.7038  1.9140
##
## Random effects:

```

```

## Groups      Name      Variance Std.Dev.
## Country_Born (Intercept) 20.77  4.557
## Residual      570.18  23.878
## Number of obs: 1871, groups: Country_Born, 13
##
## Fixed effects:
##      Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)  62.823      1.490 10.849  42.17 2.23e-13 ***
## HPP_9        -2.449      1.913 10.704  -1.28  0.228
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## HPP_9 0.240

m_Sex_Freq_2 <- lmer(Sex_Freq ~ HPP_9 + GINI + GDP + (1|Country_Born), data=db)
summary(m_Sex_Freq_2)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Sex_Freq ~ HPP_9 + GINI + GDP + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 17197.1
##
## Scaled residuals:
##      Min      1Q  Median      3Q      Max
## -2.8643 -0.6241  0.1296  0.6951  1.9076
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## Country_Born (Intercept) 19.32  4.396
## Residual      570.27  23.880
## Number of obs: 1871, groups: Country_Born, 13
##
## Fixed effects:
##      Estimate Std. Error    df t value Pr(>|t|)
## (Intercept) 52.877150  9.295473 10.749651  5.688 0.000153 ***
## HPP_9        -4.446613  3.129108  8.248730  -1.421 0.191984
## GINI          0.256854  0.177313  9.459660  1.449 0.179773
## GDP          -0.006388  0.147674  9.669816  -0.043 0.966375
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) HPP_9  GINI
## HPP_9 -0.158
## GINI  -0.886 -0.182
## GDP   -0.675  0.696  0.291

m_Sex_Freq_3 <- lmer(Sex_Freq ~ HPP_9 + GINI + GDP + Sex + Age + SRA +
  RelStatus + (1|Country_Born), data=db)
summary(m_Sex_Freq_3)

```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Sex_Freq ~ HPP_9 + GINI + GDP + Sex + Age + SRA + RelStatus +
## (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 17007.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.1619 -0.5833  0.1027  0.7246  2.1157
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## Country_Born (Intercept) 28.24    5.314
## Residual                560.65   23.678
## Number of obs: 1854, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  37.67314   11.01660   11.84344   3.420  0.00517 **
## HPP_9        -4.30669    3.64168    8.47751  -1.183  0.26907
## GINI         0.27591    0.20439    9.32497   1.350  0.20889
## GDP          0.03218    0.17024    9.49292   0.189  0.85405
## Sex2         1.30784    1.26628  1844.26255   1.033  0.30183
## Age         -0.03880    0.05087  1842.24945  -0.763  0.44565
## SRA          2.92362    0.47110  1835.94367   6.206  6.7e-10 ***
## RelStatus2   -0.03250    1.30943  1839.52677  -0.025  0.98020
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) HPP_9  GINI   GDP    Sex2   Age    SRA
## HPP_9        -0.158
## GINI         -0.851 -0.182
## GDP          -0.646  0.703  0.276
## Sex2         -0.116  0.006  0.017 -0.002
## Age          -0.118 -0.005 -0.017 -0.031  0.128
## SRA          -0.208  0.000  0.004  0.025 -0.001 -0.044
## RelStatus2   -0.032 -0.008 -0.034 -0.016  0.040  0.088  0.080
```

Summary Table sex frequency

```
class(m_Sex_Freq_1) <- "lmerMod"
class(m_Sex_Freq_2) <- "lmerMod"
class(m_Sex_Freq_3) <- "lmerMod"
stargazer(m_Sex_Freq_1, m_Sex_Freq_2, m_Sex_Freq_3, type = "text", digits = 2,
          star.cutoffs = c(0.05, 0.01, 0.001), report = ('vc*stp'))
```

```
##
## =====
##              Dependent variable:
##              -----
##              Sex_Freq
```

```

##               (1)         (2)         (3)
## -----
## HPP_9          -2.45      -4.45      -4.31
##               (1.91)      (3.13)      (3.64)
##               t = -1.28  t = -1.42  t = -1.18
##               p = 0.21   p = 0.16   p = 0.24
##
## GINI           0.26       0.28
##               (0.18)      (0.20)
##               t = 1.45   t = 1.35
##               p = 0.15   p = 0.18
##
## GDP            -0.01      0.03
##               (0.15)      (0.17)
##               t = -0.04  t = 0.19
##               p = 0.97   p = 0.86
##
## Sex2           1.31
##               (1.27)
##               t = 1.03
##               p = 0.31
##
## Age            -0.04
##               (0.05)
##               t = -0.76
##               p = 0.45
##
## SRA            2.92***
##               (0.47)
##               t = 6.21
##               p = 0.00
##
## RelStatus2     -0.03
##               (1.31)
##               t = -0.02
##               p = 0.99
##
## Constant       62.82***   52.88***   37.67***
##               (1.49)      (9.30)      (11.02)
##               t = 42.17   t = 5.69   t = 3.42
##               p = 0.00   p = 0.00   p = 0.001
##
## -----
## Observations    1,871      1,871      1,854
## Log Likelihood  -8,597.88 -8,598.57 -8,503.67
## Akaike Inf. Crit. 17,203.75 17,209.13 17,027.34
## Bayesian Inf. Crit. 17,225.89 17,242.34 17,082.60
## =====
## Note:           *p<0.05; **p<0.01; ***p<0.001

```

Models kissing satisfaction

```
m_Kiss_Satis_1 <- lmer(Kiss_Satis ~ HPP_9 + (1|Country_Born), data=db)
summary(m_Kiss_Satis_1)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Kiss_Satis ~ HPP_9 + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 17220.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.5482 -0.4766  0.2765  0.8655  0.9639
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## Country_Born (Intercept)    1.81      1.345
## Residual                  508.00    22.539
## Number of obs: 1899, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  79.5515     0.7332  5.6412 108.505 1.39e-10 ***
## HPP_9         0.5273     0.9144  4.2457   0.577   0.593
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## HPP_9 0.292
```

```
m_Kiss_Satis_2 <- lmer(Kiss_Satis ~ HPP_9 + GINI + GDP + (1|Country_Born), data=db)
```

```
## singular fit
```

```
# using blmer to deal with singularity
```

```
m_Kiss_Satis_2 <- blmer(Kiss_Satis ~ HPP_9 + GINI + GDP + (1|Country_Born), data=db)
summary(m_Kiss_Satis_2)
```

```
## Cov prior : Country_Born ~ wishart(df = 3.5, scale = Inf, posterior.scale = cov, common.scale = TRUE)
## Prior dev : 7.5972
##
## Linear mixed model fit by REML ['blmerMod']
## Formula: Kiss_Satis ~ HPP_9 + GINI + GDP + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 17222.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.6351 -0.4834  0.2981  0.8141  1.0420
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## Country_Born (Intercept)    3.201     1.789
```

```

## Residual                    506.836  22.513
## Number of obs: 1899, groups: Country_Born, 13
##
## Fixed effects:
##           Estimate Std. Error t value
## (Intercept) 71.74106    5.79502  12.380
## HPP_9       -1.35877    1.76694  -0.769
## GINI         0.20109    0.10596   1.898
## GDP        -0.01294    0.08967  -0.144
##
## Correlation of Fixed Effects:
##      (Intr) HPP_9  GINI
## HPP_9 -0.155
## GINI  -0.905 -0.157
## GDP   -0.715  0.677  0.379

m_Kiss_Satis_3 <- lmer(Kiss_Satis ~ HPP_9 + GINI + GDP + Sex + Age + SRA + RelStatus +
                      (1|Country_Born), data=db)

## singular fit
# using blmer to deal with singularity
m_Kiss_Satis_3 <- blmer(Kiss_Satis ~ HPP_9 + GINI + GDP + Sex + Age + SRA + RelStatus +
                      (1|Country_Born), data=db)
summary(m_Kiss_Satis_3)

## Cov prior : Country_Born ~ wishart(df = 3.5, scale = Inf, posterior.scale = cov, common.scale = TRUE)
## Prior dev : 7.5971
##
## Linear mixed model fit by REML ['blmerMod']
## Formula: Kiss_Satis ~ HPP_9 + GINI + GDP + Sex + Age + SRA + RelStatus +
##          (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 17009.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.8268 -0.5291  0.2836  0.7911  1.3744
##
## Random effects:
## Groups      Name             Variance Std.Dev.
## Country_Born (Intercept)    3.125    1.768
## Residual                   494.876   22.246
## Number of obs: 1881, groups: Country_Born, 13
##
## Fixed effects:
##           Estimate Std. Error t value
## (Intercept) 68.79631    6.38427  10.776
## HPP_9       -1.34666    1.75668  -0.767
## GINI         0.24529    0.10529   2.330
## GDP          0.02438    0.08934   0.273
## Sex2        -0.52802    1.18126  -0.447
## Age         -0.17270    0.04718  -3.660
## SRA          1.52023    0.43403   3.503

```

```
## RelStatus2 -4.01977 1.21243 -3.315
##
## Correlation of Fixed Effects:
## (Intr) HPP_9 GINI GDP Sex2 Age SRA
## HPP_9 -0.148
## GINI -0.821 -0.151
## GDP -0.650 0.678 0.382
## Sex2 -0.189 0.022 0.027 0.003
## Age -0.205 0.016 -0.019 -0.048 0.133
## SRA -0.340 -0.010 0.025 0.045 0.005 -0.037
## RelStatus2 -0.050 -0.018 -0.071 -0.026 0.036 0.105 0.083
```

Summary Table kissing satisfaction

```
class(m_Kiss_Satis_1) <- "lmerMod"
class(m_Kiss_Satis_2) <- "lmerMod"
class(m_Kiss_Satis_3) <- "lmerMod"
stargazer(m_Kiss_Satis_1, m_Kiss_Satis_2, m_Kiss_Satis_3, type = "text", digits = 2,
  star.cutoffs = c(0.05, 0.01, 0.001),report=('vc*stp'))
```

```
##
## =====
## Dependent variable:
## -----
## Kiss_Satis
## (1) (2) (3)
## -----
## HPP_9 0.53 -1.36 -1.35
## (0.91) (1.77) (1.76)
## t = 0.58 t = -0.77 t = -0.77
## p = 0.57 p = 0.45 p = 0.45
##
## GINI 0.20 0.25*
## (0.11) (0.11)
## t = 1.90 t = 2.33
## p = 0.06 p = 0.02
##
## GDP -0.01 0.02
## (0.09) (0.09)
## t = -0.14 t = 0.27
## p = 0.89 p = 0.79
##
## Sex2 -0.53
## (1.18)
## t = -0.45
## p = 0.66
##
## Age -0.17***
## (0.05)
## t = -3.66
## p = 0.0003
##
## SRA 1.52***
## (0.43)
```

```
##                                t = 3.50
##                                p = 0.0005
##
## RelStatus2                    -4.02***
##                                (1.21)
##                                t = -3.32
##                                p = 0.001
##
## Constant          79.55***  71.74***  68.80***
##                    (0.73)   (5.80)   (6.38)
##                    t = 108.51 t = 12.38 t = 10.78
##                    p = 0.00  p = 0.00  p = 0.00
##
## -----
## Observations          1,899      1,899      1,881
## Log Likelihood        -8,610.40  -8,611.14 -8,504.96
## Akaike Inf. Crit.    17,228.81  17,234.27 17,029.93
## Bayesian Inf. Crit. 17,251.00  17,267.57 17,085.32
## =====
## Note:                  *p<0.05; **p<0.01; ***p<0.001
```

Models Hugging satisfaction

```
m_Hug_Satis_1 <- lmer(Hug_Satis ~ HPP_9 + (1|Country_Born), data=db)
summary(m_Hug_Satis_1)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Hug_Satis ~ HPP_9 + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 16982.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.8601 -0.4878  0.3434  0.8104  0.9897
##
## Random effects:
## Groups      Name             Variance Std.Dev.
## Country_Born (Intercept)    2.716    1.648
## Residual                  458.419   21.411
## Number of obs: 1894, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)  80.8966    0.7679  7.7876 105.353 1.45e-13 ***
## HPP_9        -0.8078    0.9695  6.5643  -0.833   0.434
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## HPP_9 0.270
```



```
m_Hug_Satis_2 <- lmer(Hug_Satis ~ HPP_9 + GINI + GDP + (1|Country_Born), data=db)
summary(m_Hug_Satis_2)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Hug_Satis ~ HPP_9 + GINI + GDP + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 16985.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.8605 -0.4832  0.3448  0.8121  1.0985
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## Country_Born (Intercept)    2.651    1.628
## Residual                  458.016   21.401
## Number of obs: 1894, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  70.68366    5.45116  19.82728  12.967 3.85e-11 ***
## HPP_9        -0.80558    1.65067   8.83857  -0.488  0.6374
## GINI          0.18917    0.09942  14.36119   1.903  0.0773 .
## GDP           0.09081    0.08416  16.30127   1.079  0.2963
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) HPP_9  GINI
## HPP_9 -0.153
## GINI  -0.906 -0.158
## GDP   -0.717  0.675  0.384
```

```
m_Hug_Satis_3 <- lmer(Hug_Satis ~ HPP_9 + GINI + GDP + Sex + Age + SRA + RelStatus +
(1|Country_Born), data=db)
summary(m_Hug_Satis_3)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Hug_Satis ~ HPP_9 + GINI + GDP + Sex + Age + SRA + RelStatus +
##      (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 16726.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.9595 -0.4612  0.3180  0.7079  1.6734
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## Country_Born (Intercept)    6.342    2.518
```

```
## Residual 434.827 20.852
## Number of obs: 1876, groups: Country_Born, 13
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 70.5672 6.8489 22.0618 10.303 6.78e-10 ***
## HPP_9 -1.1925 2.0469 10.0547 -0.583 0.57301
## GINI 0.2612 0.1185 13.0155 2.204 0.04613 *
## GDP 0.1236 0.0997 13.9581 1.240 0.23533
## Sex2 0.5642 1.1121 1867.8151 0.507 0.61198
## Age -0.2636 0.0445 1746.1033 -5.924 3.78e-09 ***
## SRA 1.2782 0.4101 1723.3012 3.117 0.00186 **
## RelStatus2 -7.4947 1.1400 1866.0397 -6.574 6.33e-11 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) HPP_9 GINI GDP Sex2 Age SRA
## HPP_9 -0.150
## GINI -0.831 -0.166
## GDP -0.647 0.687 0.335
## Sex2 -0.167 0.017 0.023 0.003
## Age -0.172 0.002 -0.021 -0.043 0.132
## SRA -0.300 -0.004 0.019 0.040 0.006 -0.044
## RelStatus2 -0.047 -0.014 -0.056 -0.022 0.031 0.099 0.089
```

Summary Table Hugging satisfaction

```
class(m_Hug_Satis_1) <- "lmerMod"
class(m_Hug_Satis_2) <- "lmerMod"
class(m_Hug_Satis_3) <- "lmerMod"
stargazer(m_Hug_Satis_1, m_Hug_Satis_2, m_Hug_Satis_3, type = "text", digits = 2,
  star.cutoffs = c(0.05, 0.01, 0.001),report=('vc*stp'))
```

```
##
## =====
## Dependent variable:
## -----
## Hug_Satis
## (1) (2) (3)
## -----
## HPP_9 -0.81 -0.81 -1.19
## (0.97) (1.65) (2.05)
## t = -0.83 t = -0.49 t = -0.58
## p = 0.41 p = 0.63 p = 0.57
##
## GINI 0.19 0.26*
## (0.10) (0.12)
## t = 1.90 t = 2.20
## p = 0.06 p = 0.03
##
## GDP 0.09 0.12
## (0.08) (0.10)
## t = 1.08 t = 1.24
```

```
##                                p = 0.29  p = 0.22
##
## Sex2                            0.56
##                                (1.11)
##                                t = 0.51
##                                p = 0.62
##
## Age                            -0.26***
##                                (0.04)
##                                t = -5.92
##                                p = 0.00
##
## SRA                            1.28**
##                                (0.41)
##                                t = 3.12
##                                p = 0.002
##
## RelStatus2                     -7.49***
##                                (1.14)
##                                t = -6.57
##                                p = 0.00
##
## Constant                      80.90***  70.68***  70.57***
##                                (0.77)   (5.45)   (6.85)
##                                t = 105.35 t = 12.97 t = 10.30
##                                p = 0.00   p = 0.00 p = 0.00
##
## -----
## Observations                   1,894    1,894    1,876
## Log Likelihood                 -8,491.45 -8,492.59 -8,363.31
## Akaike Inf. Crit.             16,990.90 16,997.19 16,746.63
## Bayesian Inf. Crit.           17,013.09 17,030.47 16,802.00
## =====
## Note:                          *p<0.05; **p<0.01; ***p<0.001
```

Models Sex satisfaction

```
m_Sex_Satis_1 <- lmer(Sex_Satis ~ HPP_9 + (1|Country_Born), data=db)
summary(m_Sex_Satis_1)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Sex_Satis ~ HPP_9 + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 17405.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.8608 -0.7525  0.2394  0.9104  1.2734
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## Country_Born (Intercept)  8.73    2.955
```

```

## Residual                687.11    26.213
## Number of obs: 1857, groups: Country_Born, 13
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  71.1731     1.1463 10.4465  62.091 9.29e-15 ***
## HPP_9        -0.8093     1.4634  9.7494  -0.553   0.593
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## HPP_9 0.253

m_Sex_Satis_2 <- lmer(Sex_Satis ~ HPP_9 + GINI + GDP + (1|Country_Born), data=db)
summary(m_Sex_Satis_2)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Sex_Satis ~ HPP_9 + GINI + GDP + (1 | Country_Born)
## Data: db
##
## REML criterion at convergence: 17404.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.8745 -0.7230  0.2344  0.9028  1.2618
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## Country_Born (Intercept)  2.186   1.479
## Residual                687.580  26.222
## Number of obs: 1857, groups: Country_Born, 13
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) 61.90001     6.35380 14.63702   9.742 8.74e-08 ***
## HPP_9        -3.80320     1.83459  4.98673  -2.073  0.0930 .
## GINI          0.27143     0.11431 10.02704   2.375  0.0389 *
## GDP          -0.05780     0.09686 11.76330  -0.597  0.5620
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) HPP_9  GINI
## HPP_9 -0.146
## GINI  -0.913 -0.153
## GDP   -0.732  0.666  0.419

m_Sex_Satis_3 <- lmer(Sex_Satis ~ HPP_9 + GINI + GDP + Sex + Age + SRA + RelStatus +
(1|Country_Born), data=db)
summary(m_Sex_Satis_3)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]

```

```
## Formula: Sex_Satis ~ HPP_9 + GINI + GDP + Sex + Age + SRA + RelStatus +
##      (1 | Country_Born)
##      Data: db
##
## REML criterion at convergence: 17222.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.9548 -0.6874  0.2321  0.8636  1.5789
##
## Random effects:
##      Groups          Name          Variance Std.Dev.
## Country_Born (Intercept)    1.561    1.249
## Residual                    681.588   26.107
## Number of obs: 1840, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  52.28465    7.01249   26.89481   7.456 5.21e-08 ***
## HPP_9        -3.87181    1.76286    4.88616  -2.196  0.0807 .
## GINI          0.30822    0.11120   10.93000   2.772  0.0183 *
## GDP         -0.03495    0.09470   12.92555  -0.369  0.7181
## Sex2          5.53268    1.39783  1821.13421   3.958 7.85e-05 ***
## Age         -0.07660    0.05507   583.34351  -1.391  0.1648
## SRA           1.27481    0.51540  1014.59888   2.473  0.0135 *
## RelStatus2   -0.26954    1.44780  1828.69326  -0.186  0.8523
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) HPP_9  GINI   GDP    Sex2   Age    SRA
## HPP_9        -0.137
## GINI         -0.819 -0.141
## GDP          -0.658  0.664  0.435
## Sex2         -0.202  0.027  0.032  0.004
## Age          -0.229  0.043 -0.015 -0.045  0.135
## SRA          -0.361 -0.032  0.024  0.040 -0.002 -0.032
## RelStatus2   -0.051 -0.034 -0.076 -0.038  0.040  0.101  0.083
```

Summary Table Sex satisfaction

```
class(m_Sex_Satis_1) <- "lmerMod"
class(m_Sex_Satis_2) <- "lmerMod"
class(m_Sex_Satis_3) <- "lmerMod"
stargazer(m_Sex_Satis_1, m_Sex_Satis_2, m_Sex_Satis_3, type = "text", digits = 2,
          star.cutoffs = c(0.05, 0.01, 0.001), report = ('vc*stp'))
```

```
##
## =====
##              Dependent variable:
##      -----
##              Sex_Satis
##      (1)          (2)          (3)
##      -----
```

```

## HPP_9          -0.81      -3.80*    -3.87*
##                (1.46)      (1.83)      (1.76)
##              t = -0.55  t = -2.07  t = -2.20
##              p = 0.59  p = 0.04   p = 0.03
##
## GINI           0.27*      0.31**
##                (0.11)      (0.11)
##              t = 2.37   t = 2.77
##              p = 0.02   p = 0.01
##
## GDP            -0.06      -0.03
##                (0.10)      (0.09)
##              t = -0.60  t = -0.37
##              p = 0.56   p = 0.72
##
## Sex2           5.53***
##                (1.40)
##              t = 3.96
##              p = 0.0001
##
## Age            -0.08
##                (0.06)
##              t = -1.39
##              p = 0.17
##
## SRA            1.27*
##                (0.52)
##              t = 2.47
##              p = 0.02
##
## RelStatus2     -0.27
##                (1.45)
##              t = -0.19
##              p = 0.86
##
## Constant       71.17***  61.90***  52.28***
##                (1.15)   (6.35)   (7.01)
##              t = 62.09  t = 9.74  t = 7.46
##              p = 0.00  p = 0.00  p = 0.00
##
## -----
## Observations    1,857    1,857    1,840
## Log Likelihood  -8,702.76 -8,702.25 -8,611.41
## Akaike Inf. Crit. 17,413.52 17,416.50 17,242.81
## Bayesian Inf. Crit. 17,435.63 17,449.66 17,297.99
## =====
## Note:           *p<0.05; **p<0.01; ***p<0.001

```

Sex satisfaction by sex

```
t.test(Sex_Satis ~ Sex, db, conf.int=TRUE)
```

```
##
## Welch Two Sample t-test
```

```
##
## data: Sex_Satis by Sex
## t = -3.7356, df = 780.54, p-value = 0.0002009
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -8.361496 -2.600944
## sample estimates:
## mean in group 1 mean in group 2
##      67.68826      73.16948
group.CI(Sex_Satis ~ Sex, data = db, ci = 0.95)

##      Sex Sex_Satis.upper Sex_Satis.mean Sex_Satis.lower
## 1      1      70.24096      67.68826      65.13556
## 2      2      74.50707      73.16948      71.83189
cohen.d(Sex_Satis ~ Sex, data = db)

##
## Cohen's d
##
## d estimate: -0.2091386 (small)
## 95 percent confidence interval:
##      lower      upper
## -0.3123556 -0.1059216
```

Principal Component Analysis

PCA on the determinants of a good kiss

```
# Dataset with columns of interest
kiss_comp <- db_sample[,35:41]
kiss_comp <- cbind(kiss_comp,db_sample$Sex)
kiss_comp <- cbind(kiss_comp,db_sample$Country_Born)
kiss_comp <- cbind(kiss_comp,db_sample$HPP_9)
kiss_comp <- cbind(kiss_comp,db_sample$GINI)
kiss_comp <- cbind(kiss_comp,db_sample$GDP)
kiss_comp <- cbind(kiss_comp,db_sample$ID)
colnames(kiss_comp) <- c("GK_Breath","GK_Scent","GK_Taste","GK_Wet","GK_Contact",
                        "GK_Arous","GK_SynchStyle","Sex","Country","HPP_9","GINI",
                        "GDP","ID")

# Exclusion of missing data on the 7 variables of interest
for (i in 1:7) {
  kiss_comp <- kiss_comp[-which(is.na(kiss_comp[,i])),]
}

# Predictors
Sex <- as.factor(as.character(kiss_comp$Sex))
Country <- as.factor(as.character(kiss_comp$Country))
HPP_9 <- as.numeric(as.character(kiss_comp$HPP_9))
GINI <- as.numeric(as.character(kiss_comp$GINI))
GDP <- as.numeric(as.character(kiss_comp$GDP))

# Kaiser-Meyer-Olkin technique
```

```

KMO(kiss_comp[,1:7])

## Kaiser-Meyer-Olkin factor adequacy
## Call: KMO(r = kiss_comp[, 1:7])
## Overall MSA = 0.75
## MSA for each item =
##      GK_Breath      GK_Scent      GK_Taste      GK_Wet      GK_Contact
##      0.74          0.71          0.79          0.80          0.72
##      GK_Arous GK_SynchStyle
##      0.71          0.82

# Bartlett's Test of Sphericity
bart_spher(kiss_comp[,1:7])

## Bartlett's Test of Sphericity
##
## Call: bart_spher(x = kiss_comp[, 1:7])
##
##      X2 = 3204.529
##      df = 21
## p-value < 2.22e-16

# PCA
ncomp <- 2
kiss_pca <- prcomp(kiss_comp[,1:7], center=T, scale=T)
# Results
print(kiss_pca)

## Standard deviations (1, .., p=7):
## [1] 1.6496730 1.1200056 0.9071587 0.8533115 0.7610522 0.6920394 0.6441816
##
## Rotation (n x k) = (7 x 7):
##
##      PC1      PC2      PC3      PC4      PC5
## GK_Breath  0.3706439 -0.4849616  0.03070620 -0.1499499  0.57552472
## GK_Scent   0.4176871 -0.4440279  0.11305754 -0.1193451 -0.01760598
## GK_Taste   0.4260376 -0.2841717 -0.02296307  0.2049687 -0.72037896
## GK_Wet     0.3524492  0.2278360 -0.47127072  0.7048798  0.21503909
## GK_Contact 0.3801467  0.3953525  0.45766694  0.1008459  0.26492302
## GK_Arous   0.3717342  0.4399418  0.38748245 -0.2042170 -0.17979989
## GK_SynchStyle 0.3158073 0.2917812 -0.63564931 -0.6103520 -0.02759481
##      PC6      PC7
## GK_Breath  0.373608818  0.36495928
## GK_Scent   -0.340393859 -0.69654397
## GK_Taste   -0.007702767  0.41975436
## GK_Wet     0.133754499 -0.20195697
## GK_Contact -0.564109067  0.30191502
## GK_Arous   0.615341509 -0.25582117
## GK_SynchStyle -0.172437410  0.08974262

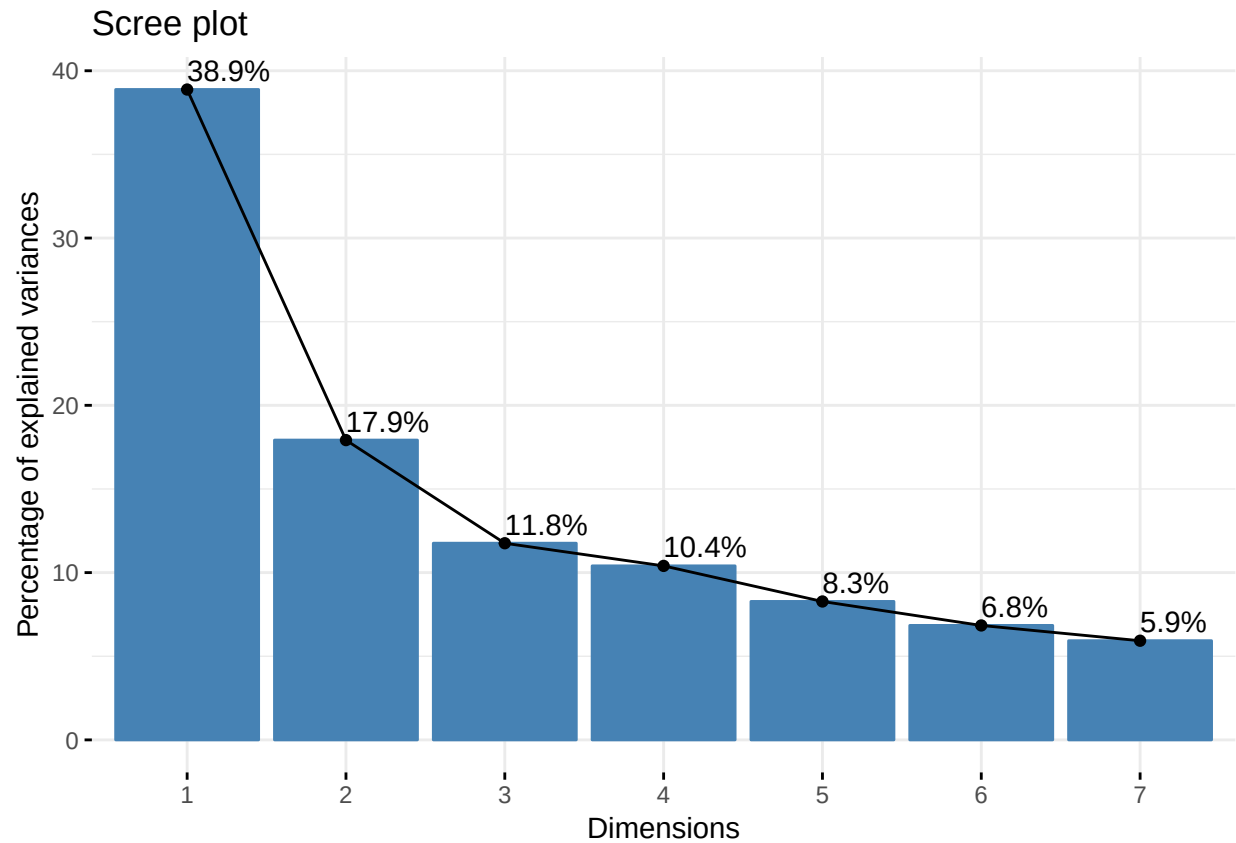
summary(kiss_pca)

## Importance of components:
##      PC1      PC2      PC3      PC4      PC5      PC6      PC7
## Standard deviation  1.6497 1.1200 0.9072 0.8533 0.76105 0.69204 0.64418
## Proportion of Variance 0.3888 0.1792 0.1176 0.1040 0.08274 0.06842 0.05928
## Cumulative Proportion 0.3888 0.5680 0.6855 0.7896 0.87230 0.94072 1.00000

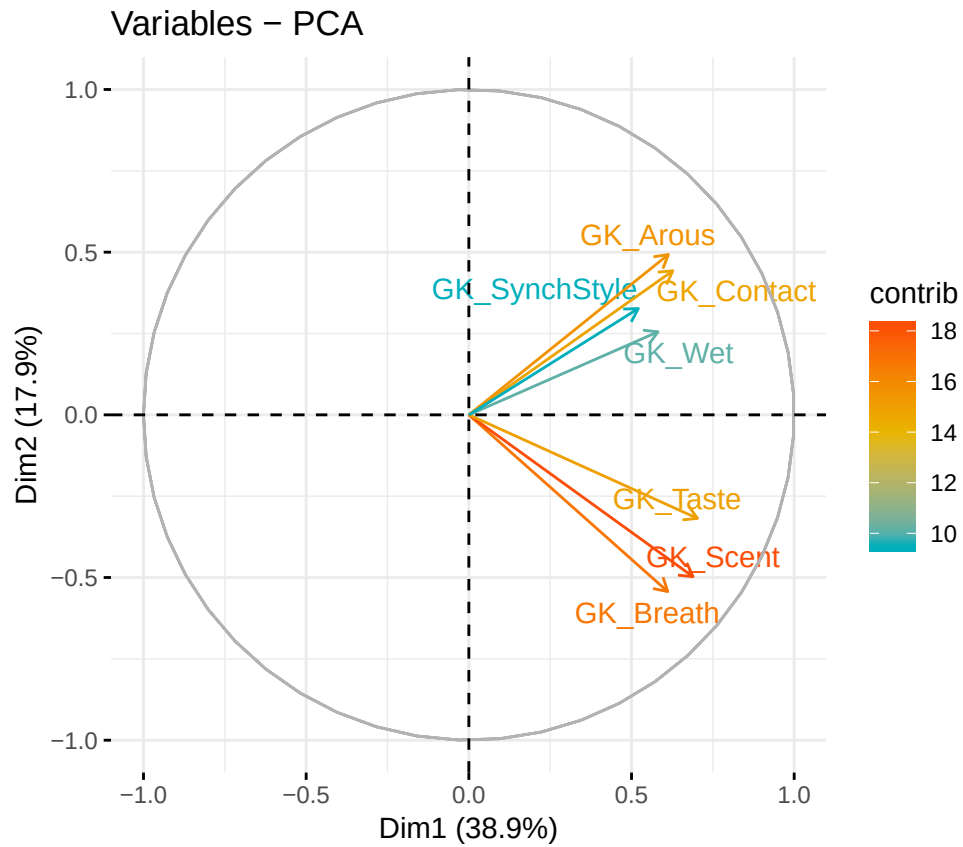
```



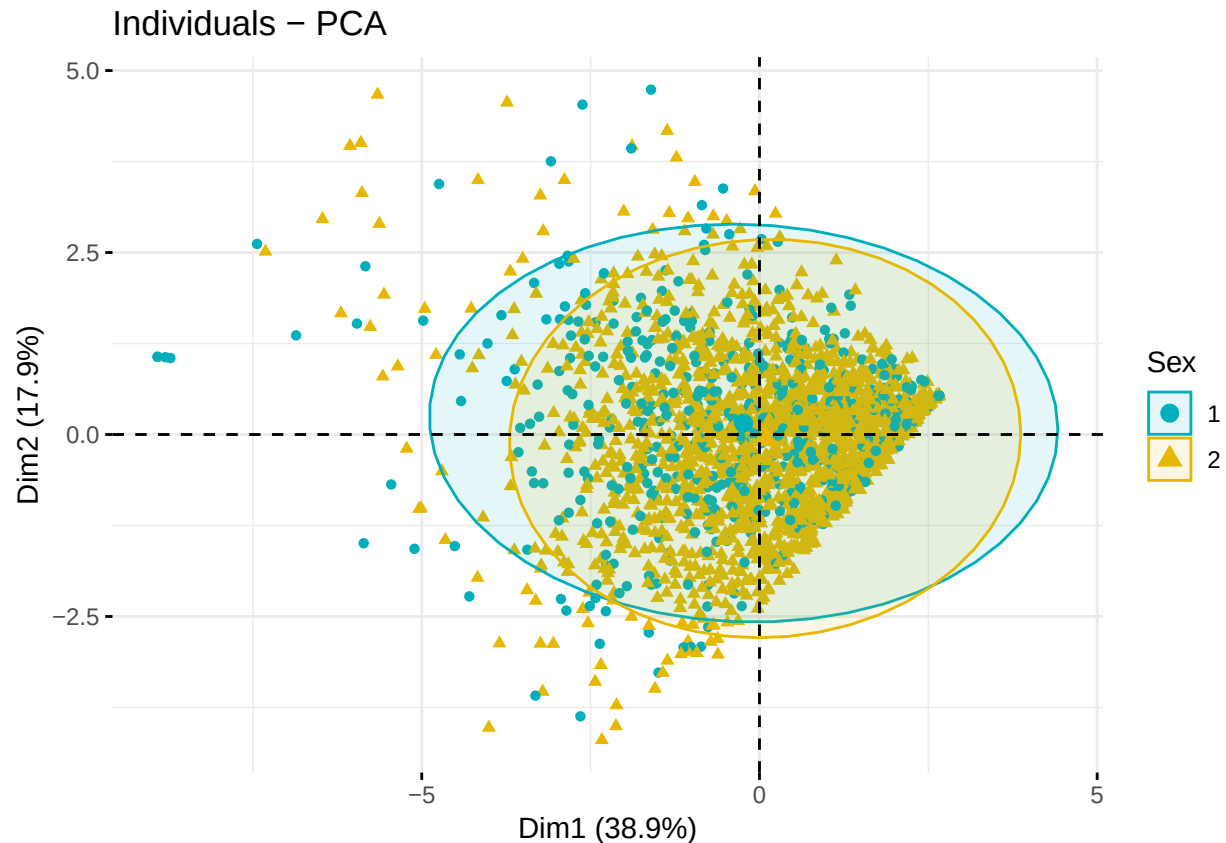
```
# Barplot of the explained variances
fviz_screepplot(kiss_pca, addlabels = TRUE)
```



```
# Graph of variables - colors using their contributions
fviz_pca_var(kiss_pca, col.var="contrib",
  gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
  repel = TRUE # Avoid text overlapping
)
```



```
# By sex
fviz_pca_ind(kiss_pca,
  geom.ind = "point",
  col.ind = Sex, # colors by groups
  palette = c("#00AFBB", "#E7B800"),
  addEllipses = TRUE, # Concentration
  legend.title = "Sex"
)
```



Varimax rotation

```
# Individuals' scores (using varimax rotation)
kiss_pca_rotated <- principal(kiss_comp[,1:7], nfactors=ncomp, rotate="varimax",
                             normalize=F, eps=1e-14)
kiss_pca_rotated

## Principal Components Analysis
## Call: principal(r = kiss_comp[, 1:7], nfactors = ncomp, rotate = "varimax",
##      normalize = F, eps = 1e-14)
## Standardized loadings (pattern matrix) based upon correlation matrix
##           RC2  RC1  h2  u2  com
## GK_Breath   0.82 0.04 0.67 0.33 1.0
## GK_Scent    0.84 0.12 0.72 0.28 1.0
## GK_Taste     0.73 0.26 0.60 0.40 1.3
## GK_Wet       0.24 0.59 0.40 0.60 1.3
## GK_Contact   0.14 0.75 0.59 0.41 1.1
## GK_Arous     0.10 0.78 0.62 0.38 1.0
## GK_SynchStyle 0.15 0.60 0.38 0.62 1.1
##
##           RC2  RC1
## SS loadings    2.01 1.97
## Proportion Var  0.29 0.28
## Cumulative Var  0.29 0.57
## Proportion Explained 0.51 0.49
## Cumulative Proportion 0.51 1.00
```

```
##
## Mean item complexity = 1.1
## Test of the hypothesis that 2 components are sufficient.
##
## The root mean square of the residuals (RMSR) is 0.11
## with the empirical chi square 1192.23 with prob < 4.6e-252
##
## Fit based upon off diagonal values = 0.86
GK_rotated_PCs <- kiss_pca_rotated$scores
GK_rotated_PCs[1:5,]

##          RC2          RC1
## 1  0.003305789 0.2352164
## 2  0.910862993 0.3562293
## 3 -0.186638851 1.0597504
## 4 -0.559737772 0.7905664
## 5  0.688094111 0.8850645

# Table with the first 2 dimensions for each individual
GK_rotated_PCs <- cbind(kiss_comp$ID, GK_rotated_PCs[,1:2])
colnames(GK_rotated_PCs) <- c("ID", "GK_rotated_dim2", "GK_rotated_dim1")
# Merging with db_sample
db_sample_2 <- merge(db_sample, GK_rotated_PCs, by = "ID", all.x = TRUE)
```

Excluding participants from the new dataset

```
db_2 <- subset(db_sample_2, db_sample_2$Ex_Cross==0)
db_2$Country_Born <- droplevels(db_2$Country_Born)
db_2$Country_Res <- droplevels(db_2$Country_Res)
```

Models with variables from the PCA

Models dim 1 (“technique”) of a good kiss

```
m_GK_rotated_dim1_1 <- lmer(GK_rotated_dim1 ~ HPP_9 + (1|Country_Born), data=db_2)
summary(m_GK_rotated_dim1_1)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: GK_rotated_dim1 ~ HPP_9 + (1 | Country_Born)
## Data: db_2
##
## REML criterion at convergence: 5047.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.4363 -0.5868  0.1442  0.7154  2.2663
##
## Random effects:
## Groups       Name             Variance Std.Dev.
## Country_Born (Intercept) 0.008803 0.09382
## Residual                0.965808 0.98276
## Number of obs: 1795, groups: Country_Born, 13
```

```
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  0.08387    0.03948  8.73710   2.125  0.06347 .
## HPP_9        0.21959    0.05006  7.80669   4.387  0.00247 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## HPP_9 0.255

m_GK_rotated_dim1_2 <- lmer(GK_rotated_dim1 ~ HPP_9 + GINI + GDP + (1|Country_Born),
                           data=db_2)
summary(m_GK_rotated_dim1_2)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: GK_rotated_dim1 ~ HPP_9 + GINI + GDP + (1 | Country_Born)
## Data: db_2
##
## REML criterion at convergence: 5064.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.4419 -0.5935  0.1436  0.7121  2.2641
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## Country_Born (Intercept) 0.01095  0.1046
## Residual                0.96583  0.9828
## Number of obs: 1795, groups: Country_Born, 13
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  0.2083784  0.2868148 13.9062093   0.727   0.480
## HPP_9        0.1546792  0.0902675  7.7689085   1.714   0.126
## GINI        -0.0003055  0.0053119 10.7831332  -0.058   0.955
## GDP        -0.0039745  0.0044655 11.7384471  -0.890   0.391
##
## Correlation of Fixed Effects:
##      (Intr) HPP_9  GINI
## HPP_9 -0.153
## GINI  -0.900 -0.167
## GDP   -0.703  0.680  0.353

m_GK_rotated_dim1_3 <- lmer(GK_rotated_dim1 ~ HPP_9 + GINI + GDP + Sex + Age + RelStatus +
                           (1|Country_Born), data=db_2)
summary(m_GK_rotated_dim1_3)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: GK_rotated_dim1 ~ HPP_9 + GINI + GDP + Sex + Age + RelStatus +
##      (1 | Country_Born)
## Data: db_2
```

```
##
## REML criterion at convergence: 5016.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.4514 -0.5915  0.1382  0.7237  2.3146
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
##   Country_Born (Intercept) 0.01025   0.1012
##   Residual                0.95441   0.9769
## Number of obs: 1780, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)  1.198e-01  2.927e-01  1.514e+01  0.409  0.68799
## HPP_9        1.624e-01  8.885e-02  7.206e+00  1.827  0.10917
## GINI         -5.477e-04  5.235e-03  1.014e+01 -0.105  0.91872
## GDP          -4.556e-03  4.409e-03  1.109e+01 -1.033  0.32338
## Sex2         -3.206e-02  5.326e-02  1.773e+03 -0.602  0.54723
## Age          5.765e-03  2.133e-03  1.485e+03  2.702  0.00697 **
## RelStatus2   -1.038e-01  5.425e-02  1.771e+03 -1.913  0.05590 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) HPP_9  GINI   GDP    Sex2   Age
## HPP_9        -0.156
## GINI         -0.866 -0.163
## GDP          -0.672  0.681  0.358
## Sex2         -0.180  0.023  0.022  0.003
## Age          -0.207  0.012 -0.025 -0.041  0.128
## RelStatus2   -0.022 -0.024 -0.059 -0.030  0.016  0.099
```

Summary Table dim 1 (“technique”) of a good kiss

```
class(m_GK_rotated_dim1_1) <- "lmerMod"
class(m_GK_rotated_dim1_2) <- "lmerMod"
class(m_GK_rotated_dim1_3) <- "lmerMod"
stargazer(m_GK_rotated_dim1_1, m_GK_rotated_dim1_2, m_GK_rotated_dim1_3, type = "text",
          digits = 2, star.cutoffs = c(0.05, 0.01, 0.001),report=('vc*stp'))
```

```
##
## =====
##              Dependent variable:
##      -----
##              GK_rotated_dim1
##              (1)          (2)          (3)
##      -----
## HPP_9              0.22***          0.15          0.16
##                   (0.05)          (0.09)          (0.09)
##                   t = 4.39      t = 1.71      t = 1.83
##                   p = 0.0001    p = 0.09      p = 0.07
##
```

```

## GINI                -0.0003   -0.001
##                    (0.01)   (0.01)
##                    t = -0.06 t = -0.10
##                    p = 0.96  p = 0.92
##
## GDP                  -0.004    -0.005
##                    (0.004)   (0.004)
##                    t = -0.89 t = -1.03
##                    p = 0.38  p = 0.31
##
## Sex2                  -0.03
##                    (0.05)
##                    t = -0.60
##                    p = 0.55
##
## Age                   0.01**
##                    (0.002)
##                    t = 2.70
##                    p = 0.01
##
## RelStatus2           -0.10
##                    (0.05)
##                    t = -1.91
##                    p = 0.06
##
## Constant             0.08*    0.21    0.12
##                    (0.04)   (0.29) (0.29)
##                    t = 2.12  t = 0.73 t = 0.41
##                    p = 0.04  p = 0.47 p = 0.69
##
## -----
## Observations         1,795    1,795    1,780
## Log Likelihood       -2,523.71 -2,532.18 -2,508.18
## Akaike Inf. Crit.    5,055.41  5,076.36  5,034.37
## Bayesian Inf. Crit.  5,077.38  5,109.31  5,083.73
## =====
## Note:                *p<0.05; **p<0.01; ***p<0.001

```

Models dim 2 (“sensory”) of a good kiss

```

m_GK_rotated_dim2_1 <- lmer(GK_rotated_dim2 ~ HPP_9 + (1|Country_Born), data=db_2)
summary(m_GK_rotated_dim2_1)

```

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: GK_rotated_dim2 ~ HPP_9 + (1 | Country_Born)
## Data: db_2
##
## REML criterion at convergence: 5030.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.4442 -0.4825  0.2291  0.7160  1.6950
##

```

```

## Random effects:
##   Groups      Name      Variance Std.Dev.
##   Country_Born (Intercept) 0.02586  0.1608
##   Residual              0.95284  0.9761
## Number of obs: 1795, groups: Country_Born, 13
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  0.02660    0.05479 11.70413   0.485   0.636
## HPP_9        0.04135    0.07018 11.38805   0.589   0.567
##
## Correlation of Fixed Effects:
##      (Intr)
## HPP_9 0.239

m_GK_rotated_dim2_2 <- lmer(GK_rotated_dim2 ~ HPP_9 + GINI + GDP + (1|Country_Born),
                           data=db_2)
summary(m_GK_rotated_dim2_2)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: GK_rotated_dim2 ~ HPP_9 + GINI + GDP + (1 | Country_Born)
##   Data: db_2
##
## REML criterion at convergence: 5040.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.4902 -0.4725  0.2276  0.7066  1.7150
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
##   Country_Born (Intercept) 0.01037  0.1018
##   Residual              0.95336  0.9764
## Number of obs: 1795, groups: Country_Born, 13
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) -0.431129    0.282599 13.719013  -1.526   0.1498
## HPP_9        -0.101519    0.088687  7.554403  -1.145   0.2873
## GINI          0.013536    0.005228 10.584750   2.589   0.0259 *
## GDP          -0.002747    0.004397 11.557759  -0.625   0.5442
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) HPP_9  GINI
## HPP_9 -0.153
## GINI  -0.900 -0.167
## GDP   -0.704  0.680  0.356

m_GK_rotated_dim2_3 <- lmer(GK_rotated_dim2 ~ HPP_9 + GINI + GDP + Sex + Age + RelStatus +
                           (1|Country_Born), data=db_2)
summary(m_GK_rotated_dim2_3)

```



```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: GK_rotated_dim2 ~ HPP_9 + GINI + GDP + Sex + Age + RelStatus +
##      (1 | Country_Born)
##      Data: db_2
##
## REML criterion at convergence: 4925
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.7077 -0.4788  0.2089  0.6899  2.0097
##
## Random effects:
##      Groups          Name          Variance Std.Dev.
## Country_Born (Intercept) 0.01905  0.1380
## Residual                0.90469  0.9512
## Number of obs: 1780, groups: Country_Born, 13
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) -1.074e+00  3.309e-01  1.480e+01 -3.245  0.00552 **
## HPP_9        -6.939e-02  1.059e-01  9.093e+00 -0.655  0.52853
## GINI         1.337e-02  6.084e-03  1.118e+01  2.197  0.04996 *
## GDP         -3.378e-03  5.093e-03  1.174e+01 -0.663  0.52006
## Sex2         2.929e-01  5.190e-02  1.773e+03  5.643 1.94e-08 ***
## Age         1.551e-02  2.088e-03  1.712e+03  7.430 1.70e-13 ***
## RelStatus2  -4.734e-02  5.285e-02  1.770e+03 -0.896  0.37048
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) HPP_9  GINI   GDP    Sex2   Age
## HPP_9        -0.157
## GINI         -0.867 -0.173
## GDP          -0.664  0.691  0.318
## Sex2         -0.156  0.016  0.020  0.002
## Age          -0.176  0.003 -0.024 -0.036  0.128
## RelStatus2  -0.020 -0.019 -0.048 -0.025  0.016  0.096
```

Summary Table dim 2 (“sensory”) of a good kiss

```
class(m_GK_rotated_dim2_1) <- "lmerMod"
class(m_GK_rotated_dim2_2) <- "lmerMod"
class(m_GK_rotated_dim2_3) <- "lmerMod"
stargazer(m_GK_rotated_dim2_1, m_GK_rotated_dim2_2, m_GK_rotated_dim2_3, type = "text",
          digits = 2, star.cutoffs = c(0.05, 0.01, 0.001),report=('vc*stp'))
```

```
##
## =====
##              Dependent variable:
##      -----
##              GK_rotated_dim2
##              (1)          (2)          (3)
##      -----
```

```

## HPP_9          0.04      -0.10      -0.07
##              (0.07)      (0.09)      (0.11)
##              t = 0.59   t = -1.14   t = -0.66
##              p = 0.56   p = 0.26    p = 0.52
##
## GINI           0.01**      0.01*
##              (0.01)      (0.01)
##              t = 2.59    t = 2.20
##              p = 0.01    p = 0.03
##
## GDP           -0.003      -0.003
##              (0.004)      (0.01)
##              t = -0.62   t = -0.66
##              p = 0.54    p = 0.51
##
## Sex2           0.29***
##              (0.05)
##              t = 5.64
##              p = 0.0000
##
## Age           0.02***
##              (0.002)
##              t = 7.43
##              p = 0.00
##
## RelStatus2     -0.05
##              (0.05)
##              t = -0.90
##              p = 0.38
##
## Constant       0.03      -0.43      -1.07**
##              (0.05)      (0.28)      (0.33)
##              t = 0.49    t = -1.53   t = -3.24
##              p = 0.63    p = 0.13    p = 0.002
##
## -----
## Observations    1,795      1,795      1,780
## Log Likelihood  -2,515.06 -2,520.45 -2,462.49
## Akaike Inf. Crit. 5,038.12 5,052.90 4,942.98
## Bayesian Inf. Crit. 5,060.09 5,085.85 4,992.33
## =====
## Note:           *p<0.05; **p<0.01; ***p<0.001

```

Participant sex and importance of the sensory component of a good kiss

```
t.test(GK_rotated_dim2 ~ Sex, db_2, conf.int=TRUE)
```

```

##
## Welch Two Sample t-test
##
## data: GK_rotated_dim2 by Sex
## t = -4.2761, df = 770.49, p-value = 2.141e-05

```

```

## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -0.3444708 -0.1277057
## sample estimates:
## mean in group 1 mean in group 2
## -0.14946024 0.08662799
group.CI(GK_rotated_dim2 ~ Sex, data = db_2, ci = 0.95)

## Sex GK_rotated_dim2.upper GK_rotated_dim2.mean GK_rotated_dim2.lower
## 1 1 -0.05400404 -0.14946024 -0.24491644
## 2 2 0.13809800 0.08662799 0.03515799
cohen.d(GK_rotated_dim2 ~ Sex, data = db_2)

##
## Cohen's d
##
## d estimate: -0.2403011 (small)
## 95 percent confidence interval:
## lower upper
## -0.3453948 -0.1352074

```